

Neurological Emergencies

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Outline

1. Status Epilepticus
2. Acute Neuromuscular
Respiratory Failure
 - Myasthenic crisis
 - Guillain-Barré Syndrome
3. Headache
4. Spinal Cord Syndromes

Bonus slides:

COVID-19
Encephalitis
Vertigo
Traumatic Brain Injury
Neurotoxins
Pregnancy

*Stroke (ischemic/hemorrhagic), MS,
Movement disorders, Neuropathy,
ALS covered in NEUROLOGY lecture
(Dr Mina Atia)*

Status epilepticus

MCQ 1- 2026

85M with prior right MCA stroke, aspiration pneumonia, on ceftriaxone, presents with a first unprovoked bilateral tonic-clonic seizure which improved after 3 min but now has rhythmic nystagmus and blinking with depressed level of consciousness and drooling. Which of the following is NOT an appropriate treatment?

- A) Midazolam 10 mg IM + Fosphenytoin 20 mg PE/kg
- B) Lorazepam 4 mg IV + Phenytoin 20 mg/kg IV
- C) Clobazam 10 mg PO + Lacosamide 5 mg/kg IV
- D) Diazepam 20 mg PR + Levetiracetam 60 mg/kg IV

Status epilepticus

Status epilepticus (SE): ≥ 5 min clinical or electrographic seizure activity OR ≥ 2 seizures without recovery in between

Refractory status epilepticus: failure of benzodiazepine + 1 Antiseizure medication (ASM) occurs in 30% of SE patients, 30% die

Rapid management:

- **Airway:** oral airway, lateral decubitus, suction, rapid sequence intubation
- **Breathing:** bag mask, 100% O₂, saturation probe, RT
- **Circulation:** cardiac monitor, IV/IO access (≥ 2 IVs)
- Fever: antipyretics, cooling, antibiotics
- CHECK CAPILLARY GLUCOSE! → thiamine 100 mg IV, D50 (50 mL IV)

Status epilepticus: Initial Management

1 Abortive + 1 Maintenance

Abortive therapy: Treat *early* with effective doses!

- Lorazepam 0.1 mg/kg IV OR 4 mg IV over 2 min, repeat 4 mg IV after 4 min
- Midazolam 10 mg IM/buccal/intranasal
- Diazepam 0.2mg/kg PR up to 20mg

Maintenance therapy

- Phenytoin 20 mg/kg IV, max 50 mg/min
- Fosphenytoin 20 mg PE/kg* IV, max 150 mg PE/min
- Valproic acid 40 mg/kg IV, at 10 mg/kg/min
- Levetiracetam 60 mg/kg IV, max 4500 mg over 15 min
- Lacosamide 200-400mg IV over 15 min
- Phenobarbital 20 mg/kg IV, at max 50 mg/min

PE: phenytoin equivalents

Silbergleit, NEJM, 2012 (VA)
Kapur, NEJM, 2019 (ESETT)

Refractory Status Epilepticus

- ICU monitoring
- Midazolam 0.2 mg/kg IV, then ≤ 2 mg/kg/h
- Propofol 2 mg /kg IV, then ≤ 10 mg/kg/h (*Avoid in mitochondrial disease or patients on ketogenic diet)
- Ketamine 1.5 mg /kg IV, then ≤ 4 mg/kg/h

Aim for seizure cessation or burst suppression for 24-48h before tapering

Monitor for non-convulsive SE with continuous EEG

especially important if patient paralyzed for intubation

Workup after tonic-clonic seizure

CT head & neck ± CTA (tumour, pus, blood)

MRI brain (cortical lesion)

EEG

Precipitants:

Glucose ↓

BUN ↑ Cr ↑

Electrolytes (Na ↓)

Extended Electrolytes (Ca ↓, Mg ↓)

Toxicology screen +, or ASM undetectable

Troponin ↑

Ammonia ↑

Expected changes post GTC:

Glucose ↑, 6-11 mmol/L

CBC (WBC ↑), electrolytes, Cr, BUN, LFTs

Lactate ↑, 5-13 mmol/L

Phosphate ↓ < 0.79 mmol/L

Bicarbonate ↓ 14-17 mmol/L

ABG/VBG (AGMA for 2h)

CK ↑ <5 x ULN

Prolactin ↑ (PPV 74% NPV 54%)

ECG

Beta-hCG

If febrile, include CSF C&S, HSV PCR

Status epilepticus: Treat the cause

Structural

- Mass lesion
- Stroke ischemic, epidural, subarachnoid, intraparenchymal hemorrhage (arterial/venous occlusion)
- Meningitis/Encephalitis (HSV 1>>2): consider antibiotics or empiric acyclovir
- Traumatic brain injury
- Hypoxic ischemic encephalopathy
- Hypertensive encephalopathy (posterior reversible encephalopathy syndrome PRES)
- Autoimmune encephalitis

Metabolic

- Hypoglycemia: thiamine 100 mg, dextrose 50% 50mL IV
- Low ASM levels: Restart missed ASM
- Withdrawal (EtOH, benzodiazepine, barbiturate), CIWA protocol
- Toxicity (bupropion, imipenem, penicillins, cephalosporins, clozapine, ifosfamide, methotrexate)
- Correct electrolyte disturbances (Na ↓, Ca ↓, Mg ↓ e.g. eclampsia)
- Renal/hepatic dysfunction

Definition of Epilepsy

≥2 unprovoked seizures >24h apart

OR

1 unprovoked seizure with >60% recurrence risk

OR

Epilepsy syndrome

ILAE 2025 Epilepsy Classification

	Focal	Generalized	Unknown whether focal or generalized
Sub-classification	Preserved vs Impaired Consciousness	Absence Tonic Clonic Other	Preserved vs Impaired Consciousness
Bilateral tonic-clonic seizures	Focal to bilateral tonic clonic	Generalized tonic-clonic	Bilateral tonic-clonic
Basic descriptors	Cognitive, Affective, Motor, Sensory, Autonomic		
EEG	Focal interictal epileptiform discharges, slowing	Generalized spike & wave	
Rx	Epilepsy surgery	ASMs Vagal nerve stimulator Ketogenic diet	Broad spectrum ASMs

	Seizure	Syncope	TIA	Migraine aura
Prodromal/ situational/ risk factors	Sleep-deprived Occurring from sleep Photic stimulation Hyperventilation Alcohol use	Lightheaded Sweating Sitting/standing Defecation/micturition Palpitations	Cerebrovascular risk factors	Family history Motion sickness Cyclic vomiting syndrome Adverse childhood experiences
During spell	Positive symptoms (i.e. presence of <i>movement, tingling, flashing lights</i>) Vocalization Déjà vu or jamais vu Epigastric sensation Head turning Tongue laceration Incontinence	Convulsions brief (few clonic jerks until <i>cerebral perfusion restored</i>) Pallor Diaphoresis	Negative symptoms (i.e. absence of <i>movement, sensation, or vision</i>) Sudden onset Maximal deficits at onset <10 min	Positive & negative symptoms Gradual migration of symptoms over 5-60 min Visual aura (scintillating scotomas) Wordfinding difficulty Paresthesias
Postdromal	Confusion Somnolence Focal neurological deficit	Alert	Alert	Associated severe, unilateral throbbing headache, photophobia, phonophobia, osmophobia, nausea/vomiting

Risk of recurrent seizure

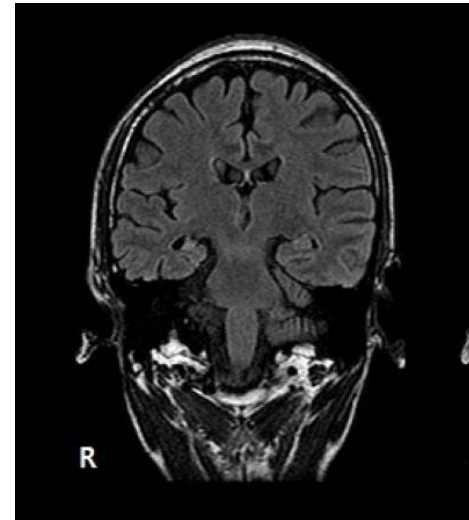
- After first unprovoked seizure, risk of recurrent seizure is ~35% at about 2 years
 - $\geq 60\%$ if abnormal EEG or MRI

EEG

- Routine Sn 20% Sp $>90\%$
- Sleep-deprived Sn 40% Sp $>90\%$

A normal EEG does NOT rule out clinical diagnosis of epilepsy!

MRI, epilepsy protocol



How to pick a first ASM

Considerations				Tolerability: “SCARED”	Sedation Cognitive Impairment Ataxia/dizziness Rash Emesis/GI upset Diplopia/visual disturbance
Efficacy	All	Valproate Lamotrigine Levetiracetam Clonazepam	Clobazam Topiramate Perampanel Rufinamide		
	Focal	Carbamazepine* Oxcarbazepine* Phenytoin* Lacosamide	Phenobarbital Eslicarbazepine* Gabapentin* Pregabalin		
	Absence	Ethosuximide Valproate			
	Myoclonic	Levetiracetam Valproate			
				Adherence	BID > TID
				Co\$t	Newer >>> Older
				Interactions	Levetiracetam renally cleared Valproate is a CYP inhibitor Others are CYP inducers
				Comorbidities	Valproate, lamotrigine, carbamazepine are mood stabilizers
				Pregnancy	Levetiracetam Lamotrigine



* Avoid in idiopathic generalized epilepsy

Adapted from Dr. Esther Bu

Favourite go-to antiseizure medications

Drug	Starting Dose / Ceiling Dose	Best for...	Not for...
Levetiracetam (PO/IV)	500 mg BID / 1500 mg BID	Hepatic dysfunction Polypharmacy Pregnancy	Psychiatric disease (hallucination, confusion, caution in elderly)
Clobazam (PO)	10mg QHS / 20 mg BID	Add-on therapy	Sedated individuals
Lacosamide (PO/IV)	50 mg BID / 200 mg BID	Status epilepticus	Bradyarrhythmia
Phenytoin (PO/IV)	100 mg TID / ~400 mg daily	Status epilepticus	Long-term maintenance
Lamotrigine (PO)	25 mg daily / 100 mg po BID	Pregnancy Outpatient setting	Status epilepticus
Valproic Acid (PO/IV)	500 mg po BID / ~4000 mg daily	Idiopathic generalized epilepsies	Pregnancy/ childbearing potential

Counselling patients with epilepsy

Safety: avoid climbing, swimming, driving

➤ **CMA Driving Guidelines:**

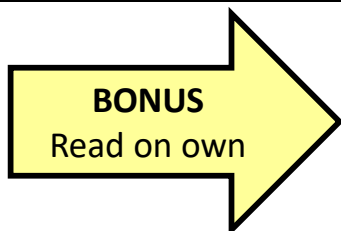
- See next slide
- Spouse/parent to take first aid course
- Sudden unexpected death in epilepsy (SUDEP)
- Medic Alert bracelet

Nonpharmacological:

- Alcohol/recreational drug cessation
- Sleep hygiene
- Trigger avoidance (photosensitivity)
- Screen for anxiety, depression, suicidal thoughts

Pharmacological:

- Affordability
- Adherence
- Interactions: beware antibiotics/OCPs/anticoagulants
- Adverse effects: “SCARED” mnemonic
 - Osteoporosis
 - Hyponatremia: carbamazepine, oxcarbazepine, eslicarbazepine
 - Psychiatric/irritability: levetiracetam
 - Stevens-Johnson syndrome: phenytoin, lamotrigine, carbamazepine, oxcarbazepine
 - PR-prolongation: lacosamide
 - Weight gain: valproate
 - Weight loss: topiramate
 - Cognitive impairment: topiramate, clobazam
 - Sedating: clobazam, phenobarbital



Driving and Seizure Disorders

CMA Driving Guide, 10th ed.

Type of Seizure	Private Driving	Commercial Driving (Truck, Bus)
Seizure – unprovoked (1 st)	No Driving 3 months Requires clearance by Neuro with EEG, imaging (MRI)	No driving commercial 12 months
Epilepsy	No driving 6 months *seizure free on medication	Resume driving 5 years seizure free (may be subject to variability by physician determination (neuro))
Medication withdrawal or change	No driving 3 months from when med was discontinued/changed	No driving 6 months from when med was discontinued/changed
Auras / simple partial seizures **beyond scope of GIM	Resume driving if: Seizures unchanged 12 months No generalized seizures Neurologist approves No impairment in LOC No head/eye deviation in seizures	Resume driving if: Seizures unchanged 3 years No generalized seizures Neurologist approves No impairment in LOC No head/eye deviation in seizures

MCQ 1- 2026

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- B) Lorazepam 4 mg IV + Phenytoin 20 mg/kg IV
- C) Clobazam 10 mg PO + Lacosamide 5 mg/kg IV Avoid choosing PO medications in this patient with reduced LOC.**
- D) Diazepam 20 mg PR + Levetiracetam 60 mg/kg IV

MCQ 2 - 2026

45F presented in the last 2 weeks with acute dysarthria, dysphagia, facial weakness, neck extensor weakness causing head drop, with fatigue in shoulder abduction. Physical examination showed tongue atrophy, neck extensor and fatigable shoulder abduction weakness, reflexes 2+ throughout. Serum was positive for anti-MuSK antibodies. Which medication would be LEAST likely to exacerbate her symptoms?

- A. Ciprofloxacin
- B. Gentamicin
- C. Rituximab
- D. Pembrolizumab
- E. Pyridostigmine

Common Pathway in Terminal Neurologic Disease

Respiratory muscle dysfunction →

Hypoventilation

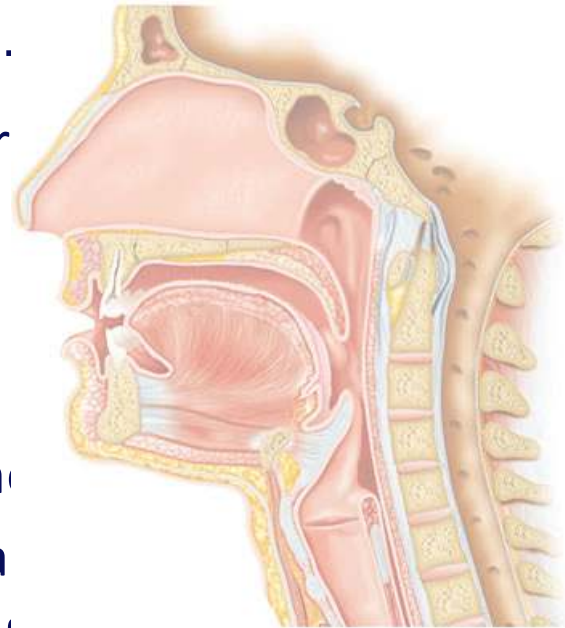
Hypercarbic respirator

OR

Bulbar muscle dysfunction

Aspiration/atelectasis

Hypoxic respiratory failure



Signs in Bulbar/Respiratory Weakness

Airway compromise

- Pooling secretions
- Snoring/stridor (below larynx)
- Stertor (above larynx)
- Inadequate gag/cough
- Paradoxical breathing

Circulation:

- Blood pressure, heart rate
- Urine output
- Temperature of extremities
- Capillary refill
- Mottling

Breathing:

- Observation
- Respiratory rate
- Auscultation
- PFTs: FVC/MIP/MEP
- ABG

Neurologic Examination

- Pattern of weakness (ocular, bulbar, respiratory, neck, proximal arm)
- Atrophy/fasciculation
- Spasticity/rigidity, hyperreflexia
- Fatigability
- Sensory level

Myasthenic Crisis

Myasthenia Gravis

Autoimmune destruction of the postsynaptic neuromuscular junction

Bimodal age of onset: young women 20s, older men 60s

Fatigable weakness primarily of

Ocular: ptosis, binocular diplopia, pupil-sparing

Bulbar: dysarthria, dysphagia, chewing fatigue, head drop

Respiratory: orthopnea

Extremities: proximal > distal weakness, and intense fatigue

Investigations

- **FVC/PFTs**
- **Serum AChR Ab** (if negative, MuSK): 80% generalized MG, 50% ocular MG
- **EMG/NCS**
 - **SFEMG**: Single-fibre EMG of frontalis, orbicularis oculi for enhanced jitter (Sn 95%, nonspecific)
 - **RNS**: Repetitive nerve stimulation for decrement (Sn 75% Sp 90%)
- **CT chest** to r/o thymoma (10% MG pts)
- **CBC, electrolytes, Cr**

- **TSH, CK**

Acute management of myasthenic crisis

**20-30-40
rule for
intubation**

Other admission orders:

SLP

PT/OT

DVT prophylaxis

bowel/bladder care

repositioning

ABCs

Elective intubation if:

$FVC < 20 \text{ mL/kg}$

$MIP \text{ } 0 \text{ to } -30 \text{ cm H}_2\text{O}$

$MEP < 40 \text{ cm H}_2\text{O}$

FVC: Forced vital capacity

MIP: Maximal inspiratory pressure, a.k.a.

NIF negative inspiratory force

MEP: Maximal expiratory pressure

Immunosuppressive agents for myasthenic crisis

Plasmapheresis (PLEX)

Preferred by consensus opinion for rapidity of action

OR

Intravenous immunoglobulin (IVIG) 2g/kg over 2-5d

Hold pyridostigmine when intubated (manage airway secretions)

High-dose prednisone *CAUTION!*****

<50% transient worsening of respiratory status in 5-10d

International consensus guidance for management of myasthenia gravis, Sanders, Neurology, 2016.

Maintenance treatment of myasthenia gravis

Symptomatic: Pyridostigmine 60mg po TID	ADVERSE EFFECTS GI upset, bronchorrhea, cholinergic crisis
Disease Modifying Therapy: Prednisone (lowest efficacious dose)	Psychosis, hyperglycemia, wt gain, bruising, osteoporosis
Azathioprine	↓ WBC, ↑ transaminases
Plasmapheresis (PLEX)	Hypogammaglobulinemia, hypocalcemia, arrhythmia, anaphylaxis
IVIg	Aseptic meningitis, thromboembolism, AKI, anaphylaxis if ↓ IgA
Eculizumab (anti-C5 / complement)	Meningococcal infection
Rituximab (anti-CD20)	Infusion reactions, opportunistic infections (PCP, PML, cryptococcal)
Efgartigimod (anti-FcRN)	URI, neutropenia
Inebilizumab (anti-CD19)	UTI, arthralgia, nasopharyngitis

Role of thymectomy in myasthenia gravis

Thymoma + (10%): Refer to thoracic surgery for thymectomy

Thymoma -: Elective thymectomy if:

- 18-65 years
- AChRAb+ generalized MG

Drugs to avoid in myasthenia

- Anesthetic agents (neuromuscular blockade)
- Antibiotics (fluoroquinolones, macrolides, aminoglycosides)
- Cardiovascular drugs (beta-blockers, procainamide, quinidine)
- Immune checkpoint inhibitors (especially anti-PD-1: nivolumab, pembrolizumab)
- Botulinum toxin
- Chloroquine, Hydroxychloroquine, quinine
- Magnesium (especially IV Mg), lithium
- Pyridostigmine ***Can paradoxically worsen MuSK-MG***
- Glucocorticoids ***CAUTION in myasthenic crisis!***
- Statins (may worsen or precipitate MG)

Guillain-Barré Syndrome

Guillain-Barré Syndrome

- Acute monophasic inflammatory demyelinating polyradiculoneuropathy causing sensory loss, ascending paralysis, and areflexia
- Wide spectrum of severity
- 1-2 : 100,000 per year
- 20% ↑ incidence for every decade above 10y
- 33% require mechanical ventilation

Risk factors:

- Antecedent infection (C. jejuni ~30%, influenza, HIV, Zika)
- Influenza (+1 case per 60,000) >>> flu shot (+1-2 cases per million vaccinations)

Acute management of GBS

**20-30-40
rule for
intubation**

Other admission orders:
SLP (50% GBS pts have
oropharyngeal weakness)

PT/OT

DVT prophylaxis
bowel/bladder care
repositioning

ABCs

Telemetry, BP, FVC q4h

Elective intubation if:

- FVC < 20mL/kg
- MIP 0 to -30 cm H₂O
- MEP < 40 cm H₂O

FVC: Forced vital capacity

MIP: Maximal inspiratory pressure, a.k.a. NIF negative
inspiratory force

MEP: Maximal expiratory pressure

Predictors of respiratory failure in Guillain-Barré Syndrome

33% of GBS patients have respiratory failure requiring mechanical ventilation

Onset to admission <7d

FVC < 60% normal predicted

Presence of facial weakness

Inability to cough

Inability to lift the head

Inability to lift the arms

Inability to stand

Dysautonomia in Guillain-Barré Syndrome

70% of GBS patients have dysautonomia

Paroxysmal hypertension (24%)

Orthostatic hypotension (19%)

Sinus tachycardia (25%)

Bradycardia, AV block

Urinary retention

Ileus

Guillain-Barré Syndrome

For nonambulatory patients within 4 weeks of symptoms,

Intravenous immunoglobulin (IVIg)

2g / kg divided over 2-5 days

OR

Plasmapheresis (PLEX)

Steroids are NOT recommended in GBS.

No role for sequential or repeat IVIg treatments

Walgaard, Lancet Neurol, 2021

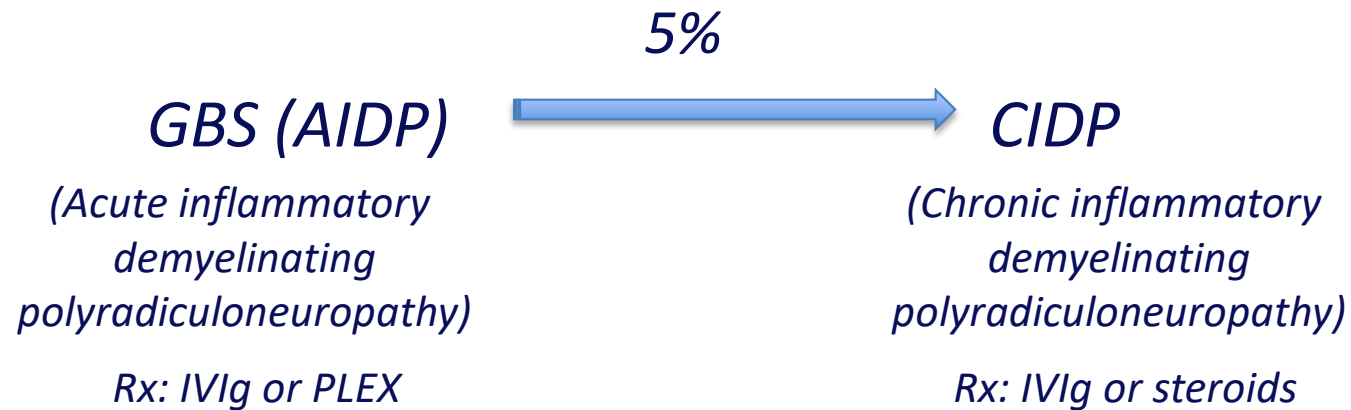
American Academy of Neurology, 2016

Investigations in Guillain-Barré Syndrome

- FVC
- Post-void residual
- MRI whole spine + Gad
 - Nerve roots & cauda equina may enhance, also r/o acute myelopathy (mimic)
- CBC, electrolytes, BUN, Cr, INR/PTT
- LP for albuminocytologic dissociation
 - Protein >0.45 + WBC <5 seen in 50% by 1st week, 75% by 3rd week.
 - Protein usually between 0.45-2
 - WBC 5-50 in 15% of GBS cases
 - If $\uparrow$ WBC, check HIV status.
- EMG/NCS to look for absent F waves, conduction blocks
 - Most sensitive 2 weeks post-symptom onset
- $\pm$ Anti-Gq1b antibodies (Miller Fisher syndrome)

Natural History of GBS

GBS is mostly monophasic,
85% walking by 1 year



When to suspect an emergent neuropathy

Typical pattern:

Distal symmetric polyneuropathy

1. Paresthesias migrate from feet to legs, leaving wake of numbness (cardboard)
2. Loss of ankle reflexes
3. ↓ sweating in feet
4. Atrophy of extensor digitorum brevis
5. Toe flexor/extensor weakness
6. Paresthesias at knees → fingertips

Red flags:

- Significant asymmetry
- Acute onset
- Early motor involvement
- Significant autonomic involvement
(orthostasis, sexual dysfunction, gastroparesis)



MCQ 2 - 2026

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- A. Ciprofloxacin can worsen neuromuscular transmission
- B. Gentamicin can worsen neuromuscular transmission
- C. Rituximab is the preferred immunotherapy for MuSK-MG**
- D. Pembrolizumab is an immune checkpoint inhibitor that can induce autoimmunity
- E. Pyridostigmine: paradoxically in MuSK-MG, unlike AChR-MG, excess acetylcholine (ACh) may accelerate postsynaptic dysfunction and ACh receptor loss

Headache

MCQ 3 - 2026

51F presents with acute headache, nausea, vomiting, blurred vision. Neurologic examination shows disc edema bilaterally. CT venogram demonstrates a vessel occlusion in the superior sagittal sinus. BP 165/93, HR 95, Hb 115, MCV 75, Plt 390, INR 1.05, Cr 70

What is the BEST first step?

- A. Mechanical thrombectomy
- B. Apixaban
- C. Enoxaparin
- D. Tenecteplase
- E. Decompressive craniectomy

Headache Differential Diagnosis

Primary headaches

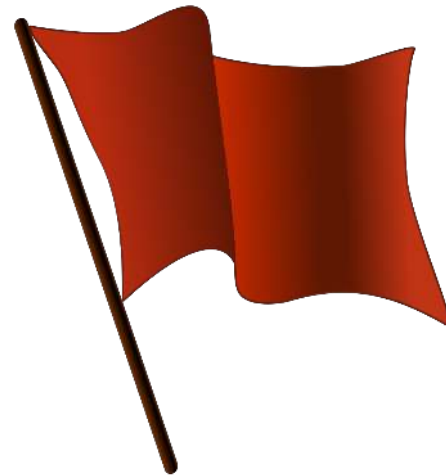
Migraine

Tension-type headache

Trigeminal autonomic cephalalgia

- Cluster headache
- Paroxysmal hemicrania
- Short-lasting unilateral neuralgiform headache attacks with conjunctival injection and tearing or cranial autonomic symptoms (SUNCT/SUNA)
- Hemicrania continua

Secondary Headaches (Dangerous)





Secondary headaches



Vascular

Ischemic stroke/venous thrombosis

Hemorrhage (ICH, SAH, SDH, EDH)

Dissection (carotid/vertebral)

Giant cell arteritis

Reversible cerebral vasoconstriction syndrome
(RCVS)

Posterior reversible encephalopathy syndrome
(PRES)

Pituitary apoplexy

Arterial-venous malformation

Space-occupying

Tumour

Hydrocephalus

Idiopathic intracranial hypertension

Infectious

Meningitis/Encephalitis/Abscess

Sinusitis/Otitis

Ophthalmologic

Acute glaucoma

Iritis

Optic neuritis



Red flags in headache: SNOOP₄



Systemic: fever, weight loss, immunosuppression (HIV, steroids, cancer, pregnancy)

Neurological symptoms/signs (meningismus, encephalopathy, papilledema)

Onset thunderclap: peaks <1min

Older > 50y

Pattern Δ , Positional, Pulsatile tinnitus, Precipitated by cough or Valsalva



Investigation of secondary headache



➤ CT/CTA/CTV head & neck

➤ ± MRI brain

“Don’t order neuroimaging or sinus imaging in patients who have a normal clinical examination, who meet diagnostic criteria for episodic migraine, and have no “red flags” for a secondary headache disorder.”

➤ CBC, electrolytes, creatinine

➤ ± ESR, CRP for GCA (Sn 85%, Sp 30%)

➤ ± D-dimer for CVT (Sn 97.8 %, Sp 84.9 %)

➤ ± blood cultures for infective endocarditis (beware all fever + stroke-like deficit)

➤ ± lumbar puncture (SAH/meningitis)

Kermani TA, Semin Arthritis Rheum, 2012.

Alons IME, BMC Neurol., 2015

Choosing Wisely, Canadian Headache Society Guidelines

Intracranial hypertension

ICP ≥ 20 mmHg (Normal ≤ 15) on opening pressure

Clinical presentation

- Headache, sometimes thunderclap
- Emesis
- Papilledema, dilated pupil, CN VI palsy
- Drowsiness

Pathophysiology

- Intracranial mass (tumour, abscess, hemorrhage)
- Cerebral edema (infarct, hypoxic-ischemic, TBI)
- Increased CSF production (choroid plexus papilloma)
- Decreased CSF absorption or obstructed outflow (post-meningitis, venous sinus thrombosis, jugular vein compression)

Investigations

- CT brain

Management

- Head of bed at 30°, maximize jugular drainage, minimize Valsalva e.g. endotracheal suctioning
- Keep CPP > 60 mmHg and $PO_2 > 60$ mmHg
- Intubate and hyperventilate patient PCO_2 26-30 mmHg
- Osmotic treatments: Hypertonic saline (3% NaCl for serum Na 145-155 mEq/L); Mannitol 20% (1 g/kg then 0.5 g/kg q6h prn); Serum osmolality > 280 mOsm/L
- Neurosurgery: External ventricular drain (EVD), Ventriculoperitoneal shunt (VP shunt), Endoscopic 3rd ventriculostomy, decompressive hemicraniectomy
- Antiseizure medication if seizures suspected
- Dexamethasone 10mg IV x 1, then 8 mg bid if vasogenic edema from mass lesion

Acute Management of Aneurysmal Subarachnoid Hemorrhage

ABCs, neuro ICU monitoring (CTA/CTP, transcranial doppler, ICP barometer, continuous EEG,)

- **Rebleeding:** ± [Antifibrinolytics (tranexamic acid), maximum 72h]
 - Reverse anticoagulation (PCC + vitamin K 10 mg IV x 1, andexanet alfa or idarucizumab)
 - Secure aneurysm (endovascular coiling vs surgical clipping)
- **Vasospasm & delayed cerebral ischemia** (day 4-14):
 - SBP < 140-160, (labetalol, nicardipine, enalapril); maintain euvolemia, euglycemia
 - Nimodipine 60 mg NG q4h for 21 days
 - Target Hb > 80
- **Elevated ICP** (previous slide):
 - **Hydrocephalus:** External ventricular drain (EVD)
 - **Hyponatremia:** SIADH: salt tabs, 3% NaCl at 10-20 mL/h, check Na q6h; AVOID water restriction!
 - AVOID hyperventilation, may worsen cerebral ischemia
- **Seizures:** ± **consider** primary prophylaxis (controversial) with Levetiracetam 500-1500 mg NG/IV BID for 1 week; treat fevers
- **Ventilator-associated pneumonia prevention:** Ceftriaxone 2g IV x 1 when intubated (PROPHY-VAP trial)
- **DVTs:** Pharmacologic thromboprophylaxis after aneurysm secured

Headache + ...

Diagnosis

Transient visual obscurations + pulsatile tinnitus + weight gain

Idiopathic intracranial hypertension

Worse standing

Spontaneous intracranial hypotension

Fever + photophobia + neck stiffness

Meningitis (infectious/SAH)

Fever + encephalopathy

Herpes simplex virus encephalitis

Fever + weight loss + scalp tenderness + jaw claudication + transient monocular vision loss + proximal myalgias + anemia

Giant cell arteritis

Recurrent thunderclap + pregnant + cannabis/decongestant use

Reversible cerebral vasoconstriction syndrome (RCVS)

Hypertension + cyclosporine/tacrolimus

Posterior reversible encephalopathy syndrome (PRES)

Seizure + hypercoagulability (hematologic malignancy, OCPs)

Cerebral venous sinus thrombosis

Neck pain + Horner syndrome

Internal carotid/vertebral artery dissection

Headache + vision loss + diplopia + hypotension + pregnant

Pituitary apoplexy

Diagnosis of migraine without aura

≥ 5 headache attacks lasting 4-72h with no better explanation

2 of:

- Unilateral location
- Pulsating quality
- Moderate or severe pain intensity
- Aggravation by or causing avoidance of routine physical activity (e.g. walking or climbing stairs)

+

1 of:

- Nausea/vomiting
- Photophobia and phonophobia

Treatment of status migrainosus

Oral

NSAIDs:

ASA 1000 mg

Diclofenac potassium 50 mg

Ibuprofen 400 mg

Naproxen 500 mg

Acetaminophen 1000 mg

Triptans:

Sumatriptan 100 mg

Rizatriptan 10 mg

Zolmitriptan 5 mg

Triptans: caution in
CAD/CVD –
vasoconstriction risk

Parenteral

Ketorolac 30-60 mg

Fluid resuscitation (100 mL/h)

Metoclopramide 10 mg q8h

Domperidone 10 mg

Magnesium sulfate 1-2 g (aura)

Dexamethasone 4-16 mg

(Dihydroergotamine 1 mg subcut)

Peripheral nerve block

Neurostimulation

Avoid opioids, barbiturates

Canadian Headache Society Guidelines

Migraine Therapy, Choosing Wisely

- Avoid opioids/barbiturates as first-line therapy
- Check for medication overuse in a headache diary before prescribing
(NSAIDS < 15, triptans < 10 days / month)
- Recommend lifestyle measures
(regular meals, sleep, trigger avoidance, stress reduction)

Chronic migraine prophylaxis

1. CGRP mAbs

Fremanezumab
Galcanezumab
Eptinezumab
Erenumab

2. Gepants

Atogepant

Anti-CGRP therapies now
strongly recommended!

CGRP: calcitonin gene-related peptide
= involved in cerebral vasodilation and pain signalling – blocking receptor can prevent or treat migraines

Gepants: small molecule CGRP antagonists used for acute treatment and prevention – minimal vasoconstriction

3. Other options

Onabotulinum toxin A
Candesartan
Propranolol
Nadolol
Amitriptyline
Butterbur
Riboflavin
Coenzyme Q10
Magnesium citrate

*Contraception: progestin or nonhormonal preferred

MCQ 3 - 2026

51F presents with acute headache, nausea, vomiting, blurred vision. Neurologic examination shows disc edema bilaterally. CT venogram demonstrates a vessel occlusion in the superior sagittal sinus. BP 165/93, HR 95, Hb 115, MCV 75, Plt 390, INR 1.05, Cr 70

What is the BEST first step?

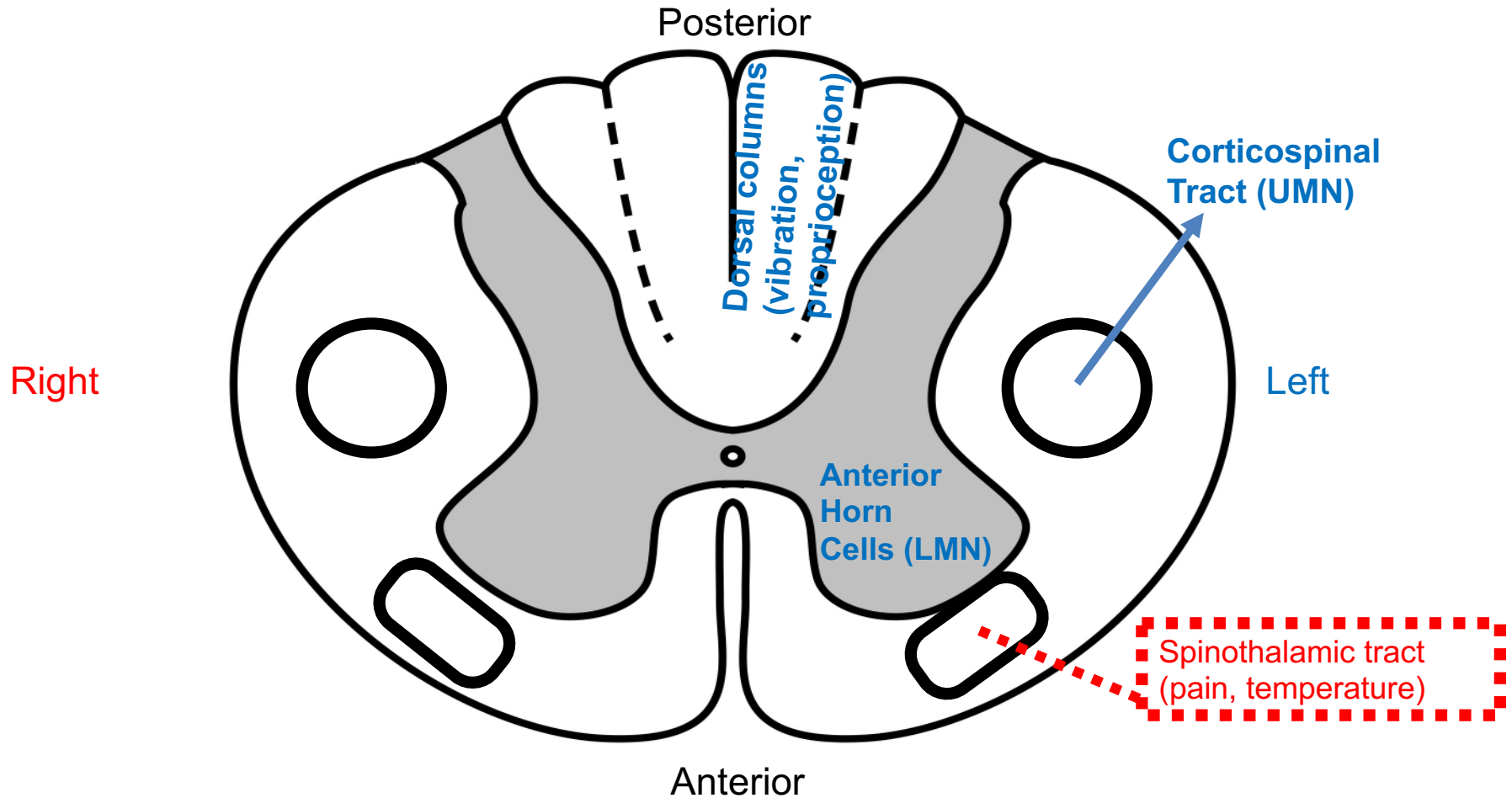
- A. Mechanical thrombectomy may be considered in select cases
- B. Apixaban: currently insufficient evidence for initial antithrombotic of choice
- C. Enoxaparin: LMWH or UFH are recommended as first-line for symptomatic CVT by Canadian Stroke Best Practices, 2025**
- D. Tenecteplase: systemic thrombolysis led to high rates of fatal hemorrhage
- E. Decompressive craniectomy may be indicated if severe mass effect

Myelopathy

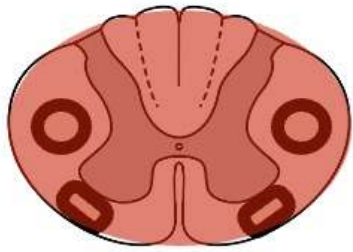
Spinal Cord Syndromes



Spinal cord cross-section

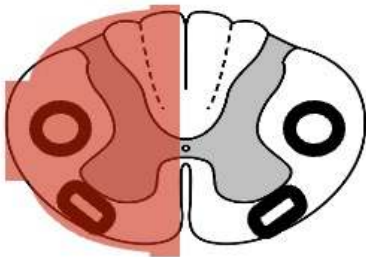


Spinal cord syndromes



Complete

Transverse
myelitis



Hemi

MS



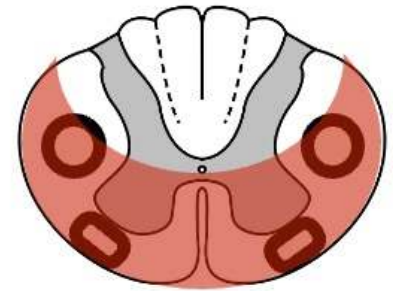
Central

Syringomyelia

Acute hyperextension
injury

Anterior

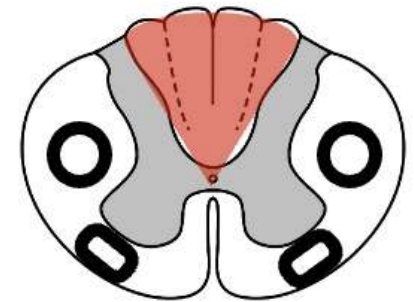
Stroke



Posterior

Syphilis

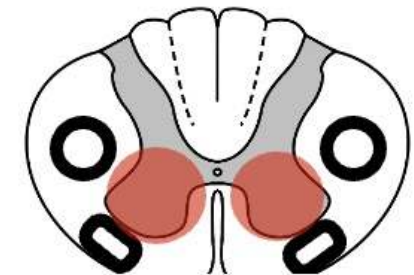
B12



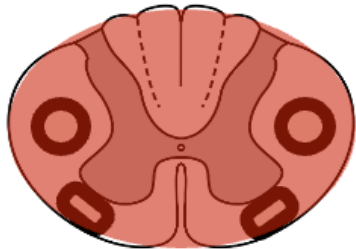
Pure Motor

West Nile Virus

ALS

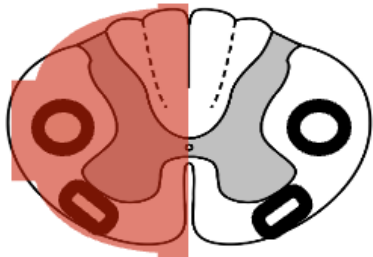


Spinal cord syndromes



Complete

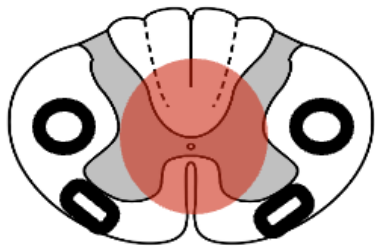
Total loss of motor and sensory below level of lesion



Hemi

Ipsilateral loss of motor/proprioception

Contralateral pain/temp loss

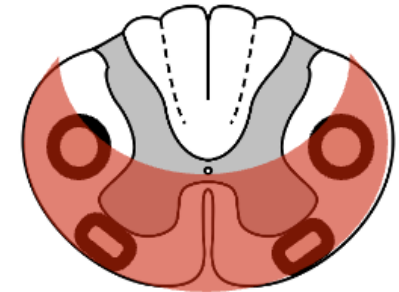


Central

Weakness UE>LE
Often sacral sparing

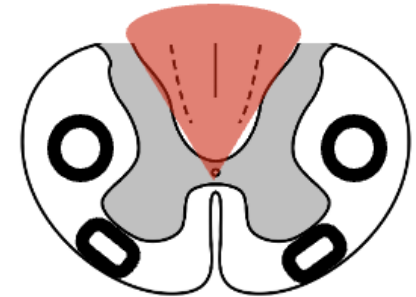
Anterior

Loss of motor, pain, temp below lesion but proprioception/vib spared



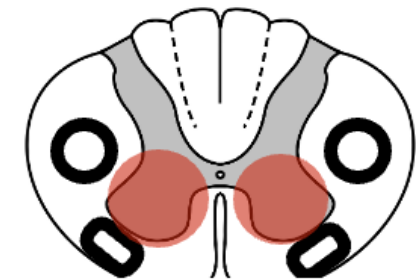
Posterior

Loss of proprioception and vibration below level



Pure Motor

Isolated motor weakness/paralysis without sensory deficits



Outline

1. Status Epilepticus
2. Acute Neuromuscular
Respiratory Failure
 - Myasthenic crisis
 - Guillain-Barré Syndrome
3. Headache
4. Spinal Cord Syndromes

Bonus slides:

COVID-19
Encephalitis
Vertigo
Traumatic Brain Injury
Neurotoxins
Pregnancy

*Stroke (ischemic/hemorrhagic), MS,
Movement disorders, Neuropathy,
ALS covered in NEUROLOGY lecture
(Dr Mina Atia)*

Neurological manifestations of COVID-19

- Anosmia (loss of smell)
- Ageusia (loss of taste)
- Acute ischemic stroke (2-6% admitted patients), large vessels in young
 - Encephalopathy
 - Encephalomyelitis
 - Headache 8%
- Myalgia 50%, Rhabdomyolysis
 - Intracerebral Hemorrhage
 - Guillain-Barré Syndrome
 - Bell's palsy

Considerations

- Protected code stroke
- CRP, D-dimer, ferritin

Special populations

- Myasthenia Gravis with COVID-19 (40% crisis)
- MS disease-modifying treatments generally safe
- Rituximab, ocrelizumab higher risk of severe COVID-19 (↑ admissions, ICU, ventilation)

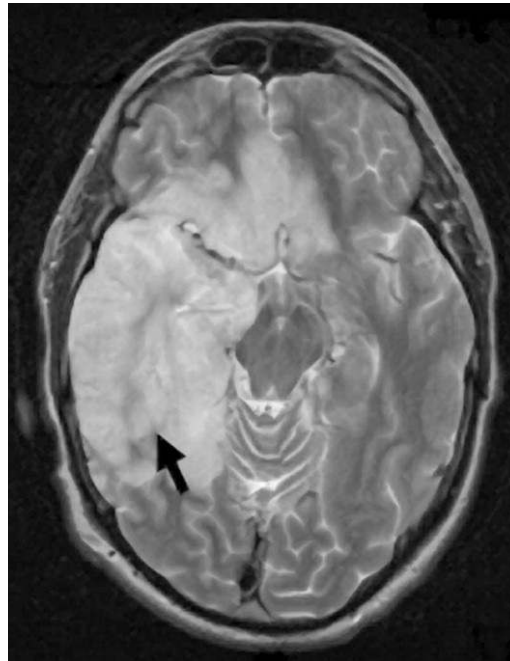
Ellul MA, Lancet, 2020
Berger JR, Neurology, 2020
Muppidi S, Lancet, 2020
8th Joint ACTRIMS/ECTRIMS 2020

BONUS MCQ 1: The obtunded octogenarian

85F with history of presents with 5 days of new confusion and drowsiness. Her husband noted that when he tried to speak with her, she appeared to stare blankly with poor comprehension. She asked repeatedly where they were going that day. She also appeared much more irritable and had decreased need for sleep at night. She comes in with an episode of tongue biting. On exam, she is febrile at 38.5°C, BP 180/100, HR 95.

She has no meningismus. She has a right complete ptosis with mydriasis and weakly opens her left eye to pain.

She has a left hemiparesis.



What is the most important treatment to initiate?

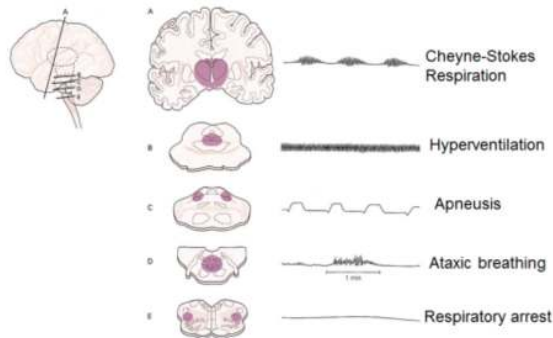
- ☐ tPA
- ☐ Prednisone
- ☐ Labetalol
- ☐ IVIg
- ✓ **Acyclovir: treat herpes encephalitis early!**

Encephalitis

Encephalopathy/coma examination

History:

- Onset (abrupt, gradual)
- Complaints (headache, depression, focal weakness, vertigo)
 - Trauma
- Medical illnesses (diabetes, renal failure, heart disease)
 - Psychiatric history
- Access to drugs (sedatives, psychotropics)



Examination:

- Vitals, ABCs, breathing pattern, glucose
- Periorbital/mastoid bruises, lateral tongue biting, embolic phenomena skin/nails, needle marks, breath odour
- MSE: meningismus, attention (backward months), memory
- CN: brainstem reflexes: Fundi, pupils, eye opening, corneals, eye movements, vestibulo-ocular, gag/cough
- Motor/sensory: Tone, motor responses (e.g. to nailbed pressure), deep tendon reflexes

Plum and Posner, *Diagnosis of Stupor and Coma*, 4th edition, 2007.
Kandel, *Principles of Neural Science*, 2000.

Acute management of encephalopathy

Emergent Treatments

- ABCs
- Thiamine, dextrose, $\pm$ Naloxone (0.4 to 2 mg IV)
- Hyperthermia: cooling blankets, antipyretics
- Consider empiric antibiotics/antivirals: ceftriaxone 2 g IV q12h, vancomycin 15mg/kg IV q8h, ampicillin 2g IV q4h, acyclovir 10 mg/kg IV q8h

Rapid investigations

- CBC, electrolytes, BUN, creatinine, glucose, ABG, osmolality, liver enzymes, NH_3 , bilirubin, albumin, INR, PTT, toxicology,
- $\pm$ TSH, B12, cortisol
- CT/CTA head and neck
- MRI brain
- EEG (Mild $\downarrow\alpha$, moderate θ , profound δ)
- Lumbar puncture

Herpes encephalitis

Most common fatal sporadic encephalitis

Pathophysiology

HSV1 reactivation along face to trigeminal ganglia around medial temporal lobes

Clinical presentation

<1 week altered mentation, aphasia, ataxia, memory impairment, or focal seizures

Investigations

- MRI brain
- Lumbar puncture: ↑ RBC, ↑ Lymphocytes, Glucose Normal
- EEG: high amplitude slow waves (delta and theta slowing), occasionally lateralized periodic discharges
- HSV CSF PCR: Sn 98% Sp >94% for 2 weeks

Prognosis

- Untreated mortality 70%
- Survivors usually disabled: 67% have residual deficits
- Associated with later development of anti-NMDAR encephalitis (psychiatric changes, dyskinesias, autonomic instability)

Autoimmune encephalitis

Overtaking infectious encephalitis as the most prevalent cause of encephalitis

Clinical presentation

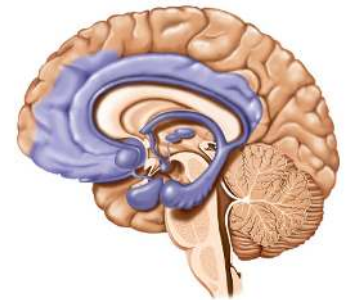
- Prodromal headache, fever, flulike illness then over weeks:
- Psychiatric changes (anxiety, irritability, psychosis)
- Movement disorders (orofacial movements, dystonia, chorea)
- Sleep disorder
- Dysautonomia (hyperthermia, hypoventilation, arrhythmias)

Investigations

- MRI brain + Gad
- EEG
- Lumbar puncture, HSV1&2/VZV, oligoclonal bands, bacterial/fungal cultures, cytology
- Serum & CSF encephalitis antibodies (London Health Sciences Centre)

Treatment

- Immunotherapy:
- Screening for malignancy: CT TAP, testicular/pelvic ultrasound, PET



BONUS MCQ 2: The vertiginous veterinarian

27F left-handed G1P1 new mother after the 20-hour labour and delivery of her 1-week old son developed 3 days of left sided neck pain which she attributed to frequent nursing and her German shepherd's yanking the leash too hard.

On her way to the chiropractor this morning, she had choked on her breakfast sandwich, began to veer to the left, and experienced vertigo with stabbing jolts of pain in the left side of her face. She missed the elevator button when reaching for it with her left hand.

On examination, HR 100, BP 130/80. She is alert. Mild dysarthria. Gaze-evoked nystagmus was noted. Left arm had hemiataxia, and gait was wide-based.



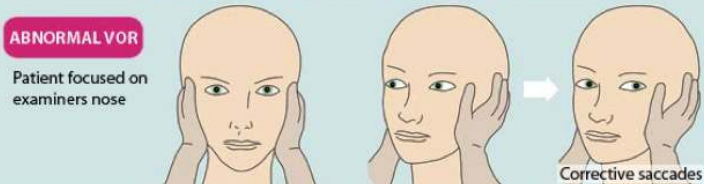
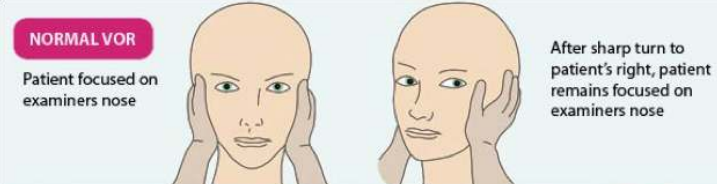
What is the most likely diagnosis?

- ☐ Status migrainosus
- ☐ Pituitary apoplexy
- ☒ **Vertebral artery dissection causing left posterior inferior cerebellar artery infarct**
- ☐ Right MCA territory infarct
- ☐ Left trigeminal neuralgia

Vertigo

Vertigo: illusory movement (rocking, tilting, moving, spinning)	Otologic	Neurologic	Medical	Psychogenic
Symptoms	Hearing loss Tinnitus Aural fullness Severe vertigo Head trauma With head postures With cough	Diplopia Dysphagia Dysarthria Hemiparesis Hemisensory loss Migraine (photophobia, phonophobia, headache) MS (prior demyelination) Stroke (vascular risks)	Syncope Shortness of breath Palpitations When standing only	Constant vertigo for months (neurologic/otologic vertigo subsides over weeks) Panic attacks
Signs	HINTS exam benign Unidirectional nystagmus Dix-Hallpike Positive Weber/Rinne test for conductive/sensorineural hearing loss	HINTS exam concerning Direction-changing nystagmus Skew deviation Horner syndrome Ataxia Focal weakness Gait instability	Orthostatic hypotension Murmur	Positive signs on functional examination: Hoover sign Distractibility Suggestibility Entrainment Waxing/waning deficits Astasia-abasia
Ancillary tests	Audiology Vestibular testing (VEMPs, electronystagmography) CT temporal bone	MRI brain	Holter Echocardiogram Electrolytes Medication review	

HINTS exam: peripheral vs central vertigo

	Peripheral (reassuring)	Central (concerning)
Head Impulse Test	Abnormal; corrective saccade with rotation  ABNORMAL VOR Patient focused on examiners nose Corrective saccades	Normal; no corrective saccade  NORMAL VOR Patient focused on examiners nose After sharp turn to patient's right, patient remains focused on examiners nose
Nystagmus	Unidirectional, horizontal	Horizontal & direction-changing; vertical; torsional
Test of Skew (vertical misalignment of eyes)	No skew deviation 	Skew deviation present 

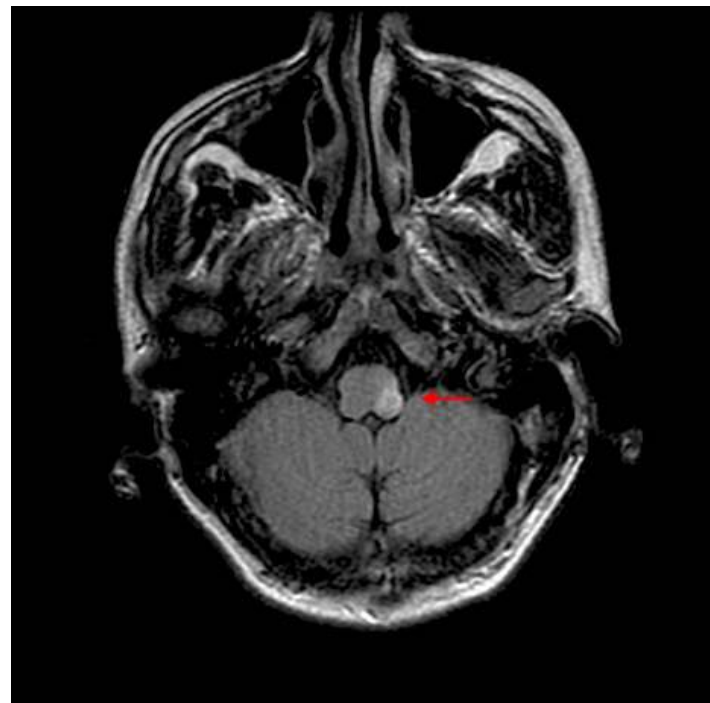
Benign HINTS exam is defined as abnormal head impulse + direction-fixed horizontal nystagmus + absent skew.

Lateral medullary syndrome (Wallenberg)

Infarction of vertebral or posterior inferior cerebellar artery (PICA)

Clinical features

- Vertigo, nausea, vomiting
- Ipsilateral Horner syndrome
- Ipsilateral face & contralateral body pain/temperature loss
- Ipsilateral cerebellar ataxia
- Ipsilateral dysphagia/dysarthria
- Hiccups
- **N.B. No limb weakness!**



BONUS MCQ 3: The photophobic pharmacist

35F was rearended in a motor vehicle crash 4 months ago and immediately afterward continues to have chronic occipital neck pain, photophobia, chronic dizziness, difficulty concentrating at work, irritability, and has been off work. Neurologic examination was normal. MRI brain was normal.

What treatment would NOT be typically indicated?

- ☐ Amitriptyline
- ☐ Mindfulness/meditation
- ✓ **Levetiracetam: no convincing benefit of this antiseizure medication compared with other mild traumatic brain injury preventive treatments**
- ☐ Duloxetine
- ☐ Venlafaxine

Traumatic brain injury

Traumatic brain injury

Concussion: Trauma-induced alteration in mental state

Mild TBI: GCS 14-15 (vast majority)

Moderate GCS 9-13

Severe GCS <9

Clinical presentation

- Confusion
- Amnesia
- Headache
- Convergence insufficiency
- Dizziness (vertigo or imbalance)
- Nausea/vomiting
- Posttraumatic seizures <5% incidence

Investigations

- SCAT6 assessment tool
- CT/MRI brain

Prognosis

- Most recover in 3 mo
- Repeated concussions rarely causes a chronic traumatic encephalopathy

Acute Management

- ABCs, ICP management
- Intubated patients: ceftriaxone 2g IV x 1

Postconcussion syndrome

Clinical presentation

- Headache
- Fatigue
- Irritability
- Photophobia, phonophobia, kinesiophobia
- Poor concentration
- Insomnia

Investigations

- Neuropsychological testing
- MRI brain
- Sleep study

Treatment

- Return to cognitive & physical activity as tolerated (NO rest)
- Treat sleep-disordered breathing
- Use chronic migraine prophylactics e.g. amitriptyline, venlafaxine, mirtazapine, candesartan

Neurotoxins

Neurologists causing medical disease

Epilepsy

- Antiepileptic drugs: Cytopenias, ↓bone mineral density, ataxia, Stevens-Johnson syndrome
- Carbamazepine: hyponatremia, mild leukopenia, lupus-like syndrome, aplastic anemia
- Phenytoin: IV phlebitis, purple glove syndrome, eosinophilia, hirsutism, gingival hyperplasia
- Valproic acid: PCOS, hyperandrogenism, insulin resistance, weight gain, tremor, hair loss, rarely hepatotoxicity, pancreatitis
- Levetiracetam: neuropsychiatric effects, irritability
- Lacosamide: ↑PR interval
- Gabapentin, pregabalin: peripheral edema

Stroke

- Anticoagulants: GI bleeds, menorrhagia, intracerebral hemorrhage

Headache

- Topiramate: acute angle closure glaucoma, nephrolithiasis, weight loss
- Botulinum toxin: UTI, dysphagia/aspiration

MS

- Methylprednisolone: delirium, psychosis, hyperglycemia, osteoporosis, shingles
- Most DMTs: lymphopenia, hepatotoxicity
- Interferons: flu-like symptoms, thyroiditis
- Fingolimod: macular edema, shingles
- Alemtuzumab: ITP, autoimmune thyroiditis, Goodpasture's syndrome

Neuromuscular

- IVIg: ACS, hemolytic anemia, aseptic meningitis, AKI
- Plasmapheresis: ↓Ca/K, ↑INR/PTT
- Pyridostigmine: GI upset, bradycardia, bronchorrhea

Internists causing neurological disease

Oncology

- PD-1/CTLA-4 inhibitors (nivolumab, ipilimumab): myasthenia gravis, myositis, hypophysitis
- CAR-T cells: cytokine release syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS)
- Microtubular (Vincristine, paclitaxel), platinum-based (cisplatin), bortezomib: polyneuropathy
- All-trans retinoic acid (other retinoids like isotretinoin): idiopathic intracranial hypertension

Rheumatology

- Anti-TNF monoclonal antibodies (adalimumab, infliximab): demyelination
- Antimetabolites (methotrexate): demyelination

Infectious Disease

- Metronidazole: corpus callosum dysmyelination (confusion, ataxia)
- Ethambutol: optic neuropathy
- Isoniazid: dorsal root ganglionopathy (from pyridoxine deficiency)
- Imipenem, penicillin, cephalosporins: seizures

Endocrinology

- Statins: immune-mediated necrotizing myopathy

Other

- Normal saline: osmotic demyelination syndrome
- Opioids: migraine, medication overuse headache
- Metoclopramide/prochlorperazine: extrapyramidal symptoms (acute dystonic reaction, parkinsonism)
- Haloperidol: neuroleptic malignant syndrome, parkinsonism

Pregnancy

Pregnancy considerations

Discuss pregnancy planning, contraception, folic acid 4-5mg/d, iron supplementation

Epilepsy

- Avoid teratogenic AEDs (valproate!)
- Drug level monitoring to correct for clearance especially 3rd trimester

Stroke [covered in Neuro lecture also]

- Highest risk peripartum to 6 weeks postpartum
- Acceptable to get CT/CTA brain before β hCG OR MRI time of flight angiography. **Avoid/no gadolinium.**
- Manage acute hypertension <160/110 mmHg
- Consult stroke neurology, maternal-fetal medicine
- Offer tPA, endovascular thrombectomy

➤ tPA doesn't cross placenta! But bleed ++ peripartum

Headache

- Secondary headaches more likely peripartum: pre-eclampsia, RCVS, SAH, PRES, CVT, pituitary apoplexy, dissection, post-LP headache
- Migraine treatments: Offer acetaminophen, sumatriptan, nerve blocks, neurostimulation

MS

- Pregnancy generally improves relapse rate, then $\uparrow$ risk in postpartum for first 3-4 months
- Offer methylprednisolone for debilitating relapses
- Glatiramer acetate and interferon beta best if planning pregnancy imminently

BONUS MCQ 1

24-year-old woman G2P2 postpartum
30 days with history of migraine
develops a sudden headache within 30s
that subsided over the next 7 days.
Medications: eletriptan. She smokes 1
joint of cannabis daily. Neuroimaging is
shown. What is the diagnosis?



What is the most likely
diagnosis?

- a) Reversible cerebral vasoconstriction syndrome
- b) Status migrainosus
- c) Pituitary apoplexy
- d) CNS vasculitis
- e) Medication overuse headache

BONUS MCQ 2

Other than diabetic neuropathy, name an important cause of neuropathy, myelopathy, cognitive dysfunction in a patient with diabetes and heart disease on aspirin, candesartan, rosuvastatin, and metformin.

- a) Vitamin E deficiency
- b) Vitamin B12 deficiency
- c) Neurosyphilis
- d) Statin-induced neuropathy
- e) Hepatosteatorsis

BONUS MCQ 3

A 20M with BMI 28 presents with numerous episodes of 3 hours of transient muscle weakness, especially after carbohydrate rich meals, alcohol, and rest after exercise. Na 140 K 2.3 Cl 110 Bicarb 25 Ca 2.2 Mg 1 Ph 1, CK 75.

What is the diagnosis?

- a) Primary hyperaldosteronism
- b) Rhabdomyolysis
- c) Deconditioning
- d) Cushing disease
- e) Hypokalemic periodic paralysis

Bonus MCQ 4 - 2026

23F has had very frequent focal impaired consciousness seizures 3 times weekly, usually in the context of alcohol intake, and she continues to have breakthrough seizures despite being on levetiracetam 1500 mg PO BID and lacosamide 200 mg PO BID for the last 2 years. EEG shows right temporal discharges. What is the LEAST appropriate next step?

- A. Add valproic acid 500 mg PO BID
- B. Refer to epilepsy monitoring unit for consideration of epilepsy surgery
- C. MRI brain, epilepsy protocol
- D. Assess medication adherence and cost
- E. Evaluate and treat her alcohol use disorder

Bonus MCQ 5 - 2026

- 20F developed epigastric rising sensation, fear, speech arrest, lip smacking movements, abnormal fumbling movements of the left hand, then rightward head and gaze deviation, followed by stiffening of the right hand, then was heard to have a loud cry and shaking movements for 2 min, after which her tongue was found with lateral laceration and she was incontinent of urine. EEG showed epileptiform discharges over the left temporal region. What is the best description of this seizure per ILAE 2025 nomenclature?
 - A. Generalized tonic-clonic seizure: incorrectly implies generalized epilepsy
 - B. Absence seizure: type of generalized seizure and does not describe semiology of impaired consciousness
 - C. Focal impaired consciousness to bilateral tonic-clonic seizure
 - D. Petit mal seizure: old nomenclature for absence seizure
 - E. Complex partial seizure: this is the old nomenclature

Bonus MCQ 6 - 2026

30F with known history of migraine presents with at least 15 days of headache in the last 3 months, of which 8 days met criteria for migraine. Which of the following does NOT have a strong recommendation for use in chronic migraine prevention?

- A. Onabotulinum toxin A injections
- B. Topiramate
- C. Fremanezumab
- D. Atogepant
- E. Propranolol

BONUS MCQ 7

50-year-old man with metastatic melanoma presents with shoulder and back pain, weakness, diplopia, ptosis, chest pain, headache, fatigue, weakness. What is the cause?

- a) Paraneoplastic myositis secondary to melanoma
- b) Aseptic meningitis secondary to intravenous immunoglobulin
- c) Hemorrhagic melanoma metastases to midbrain
- d) Ipilimumab toxicity
- e) Neuromyelitis optica

BONUS MCQ 8

A 60-year-old man with melanoma on pembrolizumab developed progressive proximal arm and leg weakness over 2 weeks, CK 6500, and ptosis. Which of the following treatments would be first-line for this condition?

- a) Methylprednisolone
- b) Intravenous immunoglobulin
- c) Plasmapheresis
- d) Stereotactic radiotherapy

BONUS MCQ 9

A 33-year-old pregnant woman G3P0A2 presents at 22 weeks gestation with a thunderclap headache and a first tonic-clonic seizure for 1 minute. CT angiogram of the brain shows a superior sagittal sinus thrombosis, and bilateral frontal intracranial hemorrhages. What is the most appropriate treatment?

- a) Tranexamic acid
- b) IV Magnesium sulphate 4g and consult OB
- c) Low molecular weight heparin: Cerebral venous thrombosis, even with secondary hemorrhage due to venous reflux, is treated with anticoagulation
- d) Lorazepam

MCQ 10- 2025

85F with prior left MCA stroke presents with a first unprovoked bilateral tonic-clonic seizure which is continuing beyond 5 min. She has a prior history of atrial fibrillation and is on apixaban. Which of the following is the most appropriate regimen?

- A) Midazolam 50 mg IM + Fosphenytoin 10 PE/kg IV
- B) Lorazepam 4 mg IV + Phenytoin 20 mg/kg IV
- C) Levetiracetam 500 mg PO BID
- D) Lamotrigine 100 mg PO BID + Clobazam 10 mg PO QHS

MCQ 11 - 2025

Which clinical parameter is LEAST concerning for impending neuromuscular respiratory failure?

- A. Forced vital capacity 900 mL
- B. Drooling and weak cough
- C. Inability to abduct the arms beyond 90° over the course of 1 week
- D. Facial paresthesias
- E. Head dropped forward to the chest

MCQ 12 - 2025

63M with obstructive sleep apnea, 1st degree heart block, diabetes mellitus, and developed fatigable ptosis, diplopia, orthopnea with mild neck extensor weakness during a COPD exacerbation. What treatment would be LEAST appropriate?

- A. Prednisone
- B. BIPAP
- C. Salbutamol
- D. Intravenous Immunoglobulin
- E. Pyridostigmine

MCQ 13 - 2025

45F presents with acute thunderclap headache and meningismus without fever and is now obtunded. CT/CTA brain shows diffuse perimesencephalic subarachnoid hemorrhage. BP 190/100, HR 90, Hb 117, MCV 80, Plt 432, INR 1.03, Na 129, Cr 100

What is the LEAST appropriate next step?

- A. Intubation
- B. Consult neurosurgery for aneurysmal clipping
- C. Nimodipine 60 mg NG q4h for 21 days
- D. Free water restriction
- E. Labetalol drip

MCQ 14 - 2025

85-year-old woman with rheumatoid arthritis and carpal tunnel syndrome develops 1 month of occipital headache, hand clumsiness, 4 falls, and new bladder urgency. What is the most appropriate investigation?

-

- A. Rheumatoid factor
- B. EMG/NCS
- C. . MRI Cervical spine
- D CK
- E. MRI Lumbar spine

Bonus answers

- 1) A - Reversible cerebral vasoconstriction syndrome presents with recurrent thunderclap headache, intracranial vessel beading, especially in peripartum period and triggered by cannabis or other vasoactive medications
- 2) B - Vitamin B12 deficiency secondary to metformin. Metformin causes poor ileal B12 absorption in up to 30% of patients. Consider monitoring or supplementing B12.
- 3) E - Hypokalemic periodic paralysis is a rare genetic channelopathy causing weakness after stress/carbohydrate meal and can be treated acutely by potassium chloride supplementation
- 4) A - A. Add valproic acid 500 mg PO BID: not the first choice due to adverse effects & teratogenicity, third ASM has diminishing likelihood of efficacy in medically refractory epilepsy
- 5) C - Focal impaired consciousness to bilateral tonic-clonic seizure
- 6) B) . Topiramate: CHS guidelines 2024 provide weak recommendation for topiramate given very low quality of evidence for efficacy in chronic migraine prevention
- 7) D) Ipilimumab toxicity: checkpoint inhibitors can cause diffuse neurologic toxicity e.g. headache, peripheral sensory neuropathy, myositis, myasthenia gravis
- 8) A) Methylprednisolone: Immune checkpoint-inhibitor induced myositis and myasthenia gravis should be treated with steroid, contrary to usual myasthenic crisis, where plasmapheresis and IVIg are preferred

Bonus answers (2)

- 9) A - Low molecular weight heparin: Cerebral venous thrombosis, even with secondary hemorrhage due to venous reflux, is treated with anticoagulation
- 10) B - Lorazepam 4 mg IV + Phenytoin 20 mg/kg IV
- 11) D - D. Facial paresthesias: bulbar/facial/neck weakness correlates more closely with respiratory weakness than paresthesias
- 12) E - Pyridostigmine: contraindicated in heart block and in myasthenic crisis where the cholinergic effects can worsen bronchorrhea and bronchospasm
- 13) D - D. Free water restriction: 3% hypertonic saline is preferred to fluid restriction to avoid volume contraction
- 14) C - C. . MRI Cervical spine: cervical myelopathy should be considered with a pattern of tetraparesis, especially in rheumatoid arthritis who are predisposed to atlantoaxial subluxation

GOOD LUCK ON YOUR EXAM!